Problem Set 3, COSC330/PHYS306
Chapters 3 and 4

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1 Chapter 3

1. **[Section 3-9, Exercise 39]** Examine the conditions indicated in Figure 3–92, and identify the faulty gates.

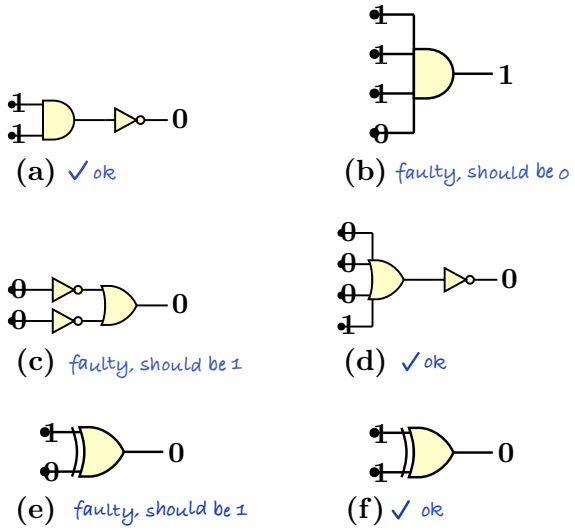


FIGURE 3–92 faulty: b, c, e

2. **[Section 3-9, Exercise 40]** Determine the faulty gates in Figure 3–93 by analyzing the timing diagrams.

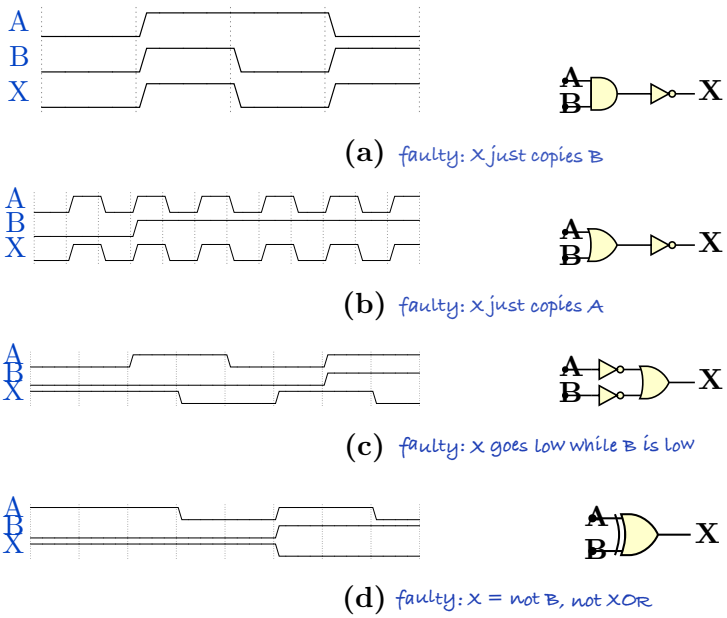


FIGURE 3–93 all 4 are faulty

2 Chapter 4

3. **[Section 4-3, Exercise 9]** Apply DeMorgan's theorems to each expression:

(a) $\overline{A+B}$ $= \overline{A}\,\overline{B}$

(b) $\overline{\overline{A}B}$ $= A + \overline{B}$

(c) $\overline{A+B+C}$ $= \overline{A}\,\overline{B}\,\overline{C}$

(d) $\overline{ABC}$ $= \overline{A} + \overline{B} + \overline{C}$

(e) $\overline{A(B+C)}$ $= \overline{A} + \overline{B}\,\overline{C}$

(f) $\overline{AB+CD}$ $= (\overline{A} + \overline{B})(\overline{C} + \overline{D})$

(g) $\overline{\overline{A}B+CD}$ $= (A + \overline{B})(\overline{C} + \overline{D})$

(h) $\overline{(A+\overline{B})(\overline{C}+D)}$ $= \overline{A}B + C\overline{D}$

4. **[Section 4-4, Exercise 13]** Write the Boolean expression for each of the logic circuits in Figure 4–57.

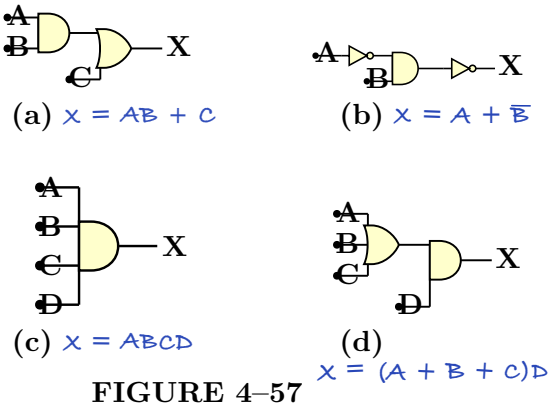


FIGURE 4–57

5. **[Section 4-4, Exercise 15]** Draw the logic circuit represented by each expression:

(a) $A(B+C)$ drawn on pg. 3

(b) $AB + ABC$

(c) $\overline{ABC} + D$

(d) $A + B(C + DB + \overline{C})$

6. **[Section 4-4, Exercise 18]** Construct a truth table for each of the following Boolean expressions:

(a) $A + B$ tables on pg. 3

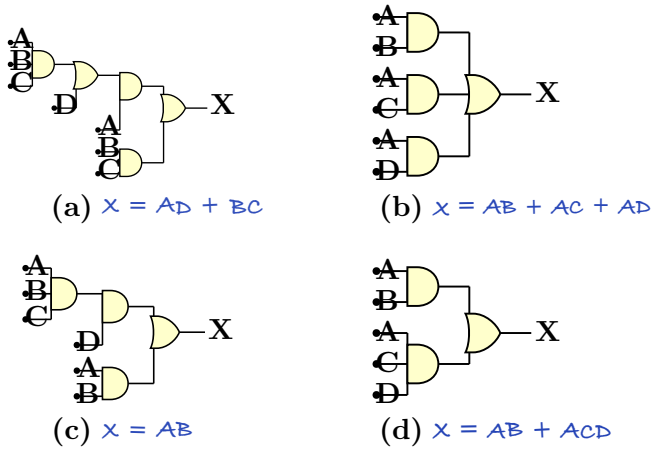
(b) AB

(c) $A + B(C + D)$ (pg. 3)

(d) $A + B + C$

(e) $AB + BC$

7. **[Section 4-4, Exercise 22]** Determine which of the logic circuits in Figure 4–59 are equivalent.



(a) $X = AD + BC$ (b) $X = AB + AC + AD$

(c) $X = AB$ (d) $X = AB + ACD$

FIGURE 4–59 none of them are equivalent

8. **[Section 4-7, Exercise 31]** Develop a truth table for each of the following standard SOP expressions:

(a) $ABC + \overline{A}BC + AB\overline{C}$ tables on right →

(b) $XYZ + X\overline{Y}Z + \overline{X}YZ + \overline{X}\overline{Y}\overline{Z} + X\overline{Y}\overline{Z}$

9. **[Section 4-7, Exercise 36]** For each truth table in Table 4–15, derive a standard SOP and a standard POS expression. on right →

ABC	X
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	0
1 1 1	1

(a)

ABC	X
0 0 0	0
0 0 1	0
0 1 0	0
0 1 1	0
1 0 0	0
1 0 1	1
1 1 0	1
1 1 1	1

(b)

$ABCD$	X
0 0 0 0	1
0 0 0 1	1
0 0 1 0	0
0 0 1 1	1
0 1 0 0	0
0 1 0 1	1
0 1 1 0	1
0 1 1 1	0
1 0 0 0	0
1 0 0 1	1
1 0 1 0	0
1 0 1 1	0
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	0

(c)

$ABCD$	X
0 0 0 0	0
0 0 0 1	0
0 0 1 0	1
0 0 1 1	0
0 1 0 0	1
0 1 0 1	1
0 1 1 0	0
0 1 1 1	1
1 0 0 0	0
1 0 0 1	0
1 0 1 0	0
1 0 1 1	1
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	1

(d)

TABLE 4–15

10. **[Section 4-9, Exercise 40]** Use a Karnaugh map to find the minimum SOP form for each expression:

(a) $\overline{A}\,\overline{B}C + A\overline{B}C + \overline{A}B\overline{C}$

(b) $AC(\overline{B} + C)$

(c) $\overline{A}(BC + B\overline{C}) + A(BC + B\overline{C})$

(d) $\overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + \overline{A}B\overline{C} + AB\overline{C}$

10) K-maps (min SOP)

(a) $X = \overline{B}C + \overline{A}B\overline{C}$

A\BC	00	01	11	10
0		1		1
1		1		

(b) $X = AC$

A\BC	00	01	11	10
0				
1		1	1	

(c) $X = B$

A\BC	00	01	11	10
0			1	1
1			1	1

(d) $X = B\overline{C} + A\overline{C} + \overline{A}\overline{B}C$

A\BC	00	01	11	10
0		1		1
1	1			1

8) truth tables

A B C	X
0 0 0	0
0 0 1	0
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	0
1 1 0	1
1 1 1	1

(a)

X Y Z	X
0 0 0	0
0 0 1	0
0 1 0	1
0 1 1	1
1 0 0	1
1 0 1	1
1 1 0	0
1 1 1	1

(b)

9) standard SOP / POS

(a) SOP $X = \overline{A}B\overline{C} + AB\overline{C} + A\overline{B}\overline{C} + ABC$

POS $(A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$

(b) SOP $X = A\overline{B}C + AB\overline{C} + ABC$

POS $(A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(A + \overline{B} + \overline{C})$
$(\overline{A} + B + C)$

(c) SOP $X = \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} + AB\overline{C}\overline{D} + ABC\overline{D}$

POS $(A + B + \overline{C} + D)(A + \overline{B} + C + D)(A + \overline{B} + \overline{C} + \overline{D})$
$(\overline{A} + B + C + D)(\overline{A} + B + \overline{C} + D)(\overline{A} + B + \overline{C} + \overline{D})$
$(\overline{A} + \overline{B} + C + \overline{D})(\overline{A} + \overline{B} + \overline{C} + D)(\overline{A} + \overline{B} + \overline{C} + \overline{D})$

(d) SOP $X = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} + \overline{A}BC\overline{D} + A\overline{B}C\overline{D} + AB\overline{C}\overline{D} + ABCD$

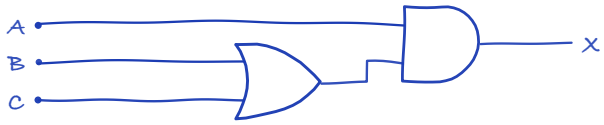
POS $(A + B + C + D)(A + B + C + \overline{D})(A + B + \overline{C} + \overline{D})$
$(A + \overline{B} + \overline{C} + D)(\overline{A} + B + C + \overline{D})(\overline{A} + B + \overline{C} + \overline{D})$
$(\overline{A} + B + \overline{C} + D)(\overline{A} + \overline{B} + C + \overline{D})(\overline{A} + \overline{B} + \overline{C} + D)$

PS 3 (continued)

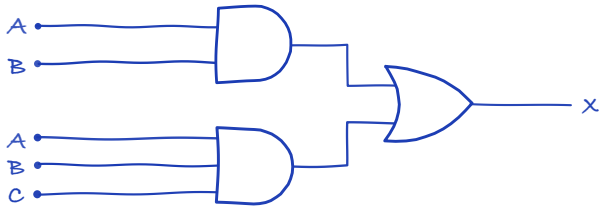
#5 circuits and #6 truth tables

5)

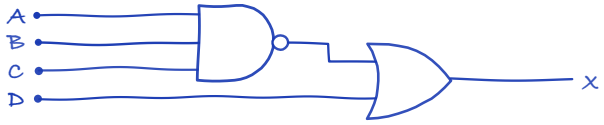
(a) $A(B + C)$



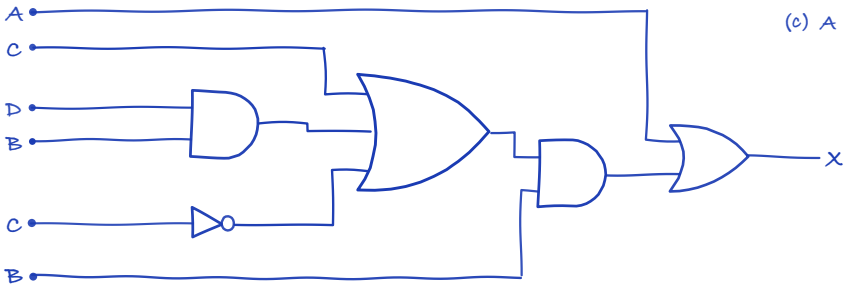
(b) $AB + ABC$



(c) $\overline{ABC} + D$



(d) $A + B(C + DB + \overline{C})$



*(b) is really just AB and (d) is just A + B, but I drew them the way they were written

6)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(a) $A + B$

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

(b) AB

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

(d) $A + B + C$

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(e) $AB + BC$

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

(c) $A + B(C + D)$